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Rapid magnetic solid-phase extraction combined with ultra-high performance liquid chromatography and quadrupole-time-of-flight mass spectrometry for analysis of thrombin binders from a crude extract and injection of *Erigeron breviscapus*†

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Thrombin, which acts as a serine protease that converts soluble fibrinogen into insoluble strands of fibrin, plays crucial roles in the blood coagulation pathway. Inhibition of thrombin leading to the decrease of thrombus formation may be used for treating cerebral ischemia and infarction. In the present study, a new approach of thrombin immobilized magnetic nanoparticles coupled with ultra-high performance liquid chromatography and quadrupole-time-of-flight mass spectrometry was established for rapidly extraction of thrombin binders from a crude extract of *Erigeron breviscapus* and its injection. Scutellarin, a model inhibitor of thrombin, was chosen to interrogate the optimal extraction conditions, including wash times, ion strength, temperature, pH, and incubation time. Meanwhile, the specificity of the approach was investigated by using denatured thrombin and the inactive compound specnuezhenide. The optimal screening conditions were as follows: wash times: 4, wash solvent: 50% acetonitrile–water, incubation time: 70 min, temperature: 37 °C, ion strength: 10 mM, pH: 6.8. Moreover, the linearity, accuracy, precision, and matrix effect of the approach were well validated. Finally, thrombin binders were extracted from the crude extract and injection of *Erigeron breviscapus* by applying the established approach, and were subsequently sent to be validated by a thrombin inhibitory assay. Eleven thrombin inhibitors were identified, and ten of which, including erigoster B, scutellarin A, erigeroside, neochlorogenic acid, chlorogenic acid, 4-caffeoylquinic acid, luteolin 7-*O*-β-D-glucuronide, 4,5-di-*O*-caffeoylquinic acid, 3,4-di-*O*-caffeoylquinic acid and 3,5-di-*O*-caffeoylquinic acid, were reported for the first time.

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Introduction

Thrombin, a potent vasoconstrictor and mitogen, is implicated as a major factor in vasospasm following subarachnoid hemorrhage. Blood from a ruptured cerebral aneurysm clots around a cerebral artery, releasing thrombin. This can induce an acute and prolonged narrowing of the blood vessel, potentially resulting in cerebral ischemia and infarction (stroke).[1,2] Therefore, inhibition of thrombin leading to the decrease of thrombus formation may be used for treating cerebral ischemia and infarction.

Erigeron breviscapus, which is a folk medicine distributed in Yunnan of China, has the function of removing blood stasis, promoting blood circulation, and activating meridians to stop pain.[3] Erigeron injections are prepared from the aqueous extracts of *Erigeron breviscapus*, which showed significant curative effects to cardiovascular and cerebral vessel diseases.[4] Phenolic acids and flavonoids are considered to be the major constituents in *Erigeron breviscapus*. Scutellarin, as a major component, had been used in clinical for the treatment of ischemic cerebrovascular diseases such as cerebral thrombosis, cerebral embolism, stroke and other sequelae caused by paralyzed patients.[5] The pharmacological effects of scutellarin included dilation of cerebral blood vessels, reducing cerebral vascular resistance, increasing cerebral blood flow, improving microcirculation, anti-thrombin and antiplatelet aggregation.[6] Therefore, it was hypothesized that other anti-thrombin agents might exist in the plant.

Magnetic nanoparticles have recently been examined as promising platforms for detection and separation applications, including protein purification,[7] ligand fishing,[8] DNA

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extraction,[9] pathogen detection[10] and cell separation.[11] The surface of magnetic nanoparticles could be easily modified and coated with antibodies, enzymes, proteins or specific ligands that enable them to bind to other biologically active compounds or receptors on the cell surface. The superparamagnetic properties facilitate the separation of the biomolecule conjugated magnetic beads from their matrix. Using magnetic nanoparticles as the support of immobilized enzymes has some advantages:[12] (i) higher specific surface area obtained for the binding of a larger amount of enzymes, (ii) lower mass transfer resistance and less fouling, and (iii) the selective separation of immobilized enzymes from a reaction mixture by the application of a magnetic field. The number of reports about application of magnetic nanoparticles is increasing year by year.

In the present, a new thrombin immobilized magnetic nanoparticles based approach was established to rapidly extraction of thrombin binders from crude extract of *Erigeron breviscapus* and its injection. Schematic diagram of the thrombin immobilized magnetic nanoparticles based solid-phase extraction approach was shown in Fig. 1. First, thrombin is immobilized on magnetic nanoparticles by amide linkage using 1-(3-dimethylaminopropyl)-3-ethylcarbodiimide (EDC) and *N*-hydroxysuccinimide (NHS) as reaction reagents. Secondly, the crude extract and injection of *Erigeron breviscapus* were incubated with the thrombin immobilized magnetic nanoparticles. Subsequently, the unbound compounds and nonspecific binding compounds were separated and discarded in the washing step; while the bound compounds were disclosed from thrombin by the organic denature reagents. Finally, the disclosed specific bound compounds were analyzed by UHPLC-Q-TOF-MS and the activities of bound compounds were validated using thrombin inhibitory assay.

Experimental

Chemicals and reagents

Thrombin (from bovine plasma, EC:3.4.21.5), 1-(3-dimethyl-??aminopropyl)-3-ethyl-carbodiimide (EDC), *N*-hydroxysuccinimide (NHS), 2-(*N*-morpholino)ethanesulfonic acid (MES) were purchased from Sigma Chemical Co. Chromogenic substrate H-D-Phe-Pip-Arg-pNA·2HCl was purchased from Beijing Adhoc International Technologies Co. Monodispersed carboxyl functionalized magnetite nanoparticles were obtained from Tianjin BaseLine ChromTech Research Centre. HPLC-grade acetonitrile was obtained from Merck. 5-Caffeoylquinic acid, 3-caffeoylquinic acid, 4-caffeoylquinic acid, 4,5-di-*O*-caffeoylquinic acid, 3,4-di-*O*-caffeoylquinic acid and 3,5-di-*O*-caffeoylquinic acid were purchased from Sichuan Weikeqi Biological Technology Co. Erigeroside, scutellarin, erigoster B, scutellarin A and luteolin 7-*O*-β-D-glucuronide were obtained from Chengdu Purechem-standard Co. Linolenic acid, linoleic acid and hexadecanoic acid were purchased from Shanghai Yuanye Bio-Technology Co. Argatroban was obtained from Dalian Meilun Shengwu Technology Co. 96-Well microtiter plates were purchased from Corning Inc. All solutions and dilutions were prepared with ultrapure water from a milli-Q water purification system.

Apparatus

The analytical apparatus included a transmission electron microscopy (FEI Tecnai G2 F20 S-TWIN), a X-ray diffractometer (XPert PRO), a vibration sample magnetometer (PPMS-9), a Fourier transform infrared spectroscopy (Bruker Optics), a ZetaMaster3000 zeta potential laser particle size analyzer (Marven), an Agilent 1100 LC-UV system with chemstation (Agilent), a Shimadzu UHPLC

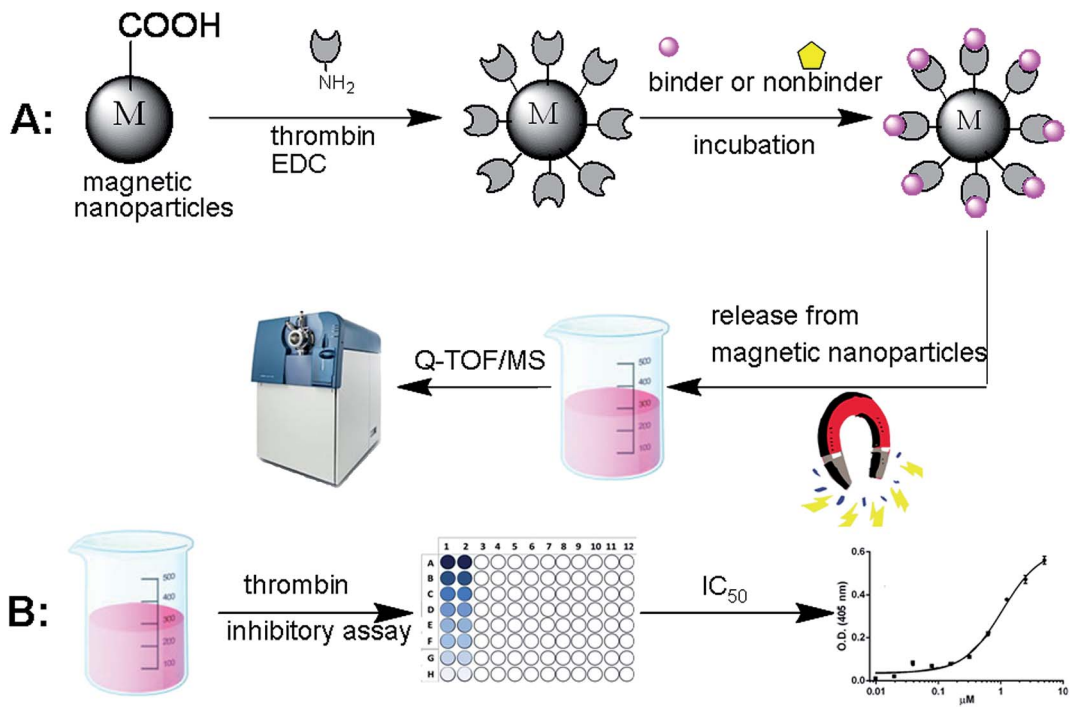


Fig. 1 Schematic diagram of the thrombin immobilized magnetic nanoparticles based solid-phase extraction approach.

system (Shimadzu), a Q-TOF 5600-plus mass spectrometer equipped with Turbo V sources and a TurboIonspray interface (AB sciex), a PB-10 Sartorious pH meter (Sartorious), a Synergy 2 multimode reader (Biotek).

Preparation of thrombin immobilized magnetic nanoparticles

Immobilization of thrombin onto the carboxylic acid-terminated magnetic nanoparticles was carried out by forming an amide bond between the carboxyl group of the magnetic nanoparticles[13] and primary amino groups of thrombin. Briefly, an aliquot of 50 μL magnetic nanoparticles was washed with 100 μL MES buffer (25 mM) for three times in a 96-well microtiter plate. An aliquot of 50 μL EDC solution (50 mg mL^{-1}, freshly prepared) and 50 μL NHS solution (50 mg mL^{-1}, freshly prepared) were added immediately to the magnetic nanoparticles. The plate was shaken in a specialized stirring mixer for 30 min at room temperature. Meanwhile, about 1 mg of thrombin was dissolved in 1 mL MES buffer (25 mM). Then, 200 μL of thrombin solution was added to the activated magnetic nanoparticles and shaken gently in the specialized stirring mixer at 4 °C. After an overnight incubation, the supernatant of the solution was separated using permanent magnet and abandoned. Subsequently, 100 μL of 0.5% (w/v) aqueous bovine serum albumin solution was added to the well to quench the reaction. The 96-well microtiter plate was shaken in the specialized stirring mixer for 15 min, after which the supernatant was discarded. Finally, the thrombin immobilized magnetic nanoparticles were stored at 4 °C for the next experiments.

Morphology characterization and protein measurement

The morphology of thrombin conjugated magnetic nanoparticles was characterized using transmission electron microscopy (TEM). Fourier transform infrared spectroscopy (FTIR) was acquired on a Bruker infrared spectrometer. Magnetization curve was measured in a vibration sample magnetometer. The crystalline structure of magnetic beads was identified using powder X-ray diffractometer. Hydrodynamic sizes and zeta potential were detected using zeta potential laser particle size analyzer.

Experiments with varying amounts of magnetic nanoparticles (25, 50, 75 and 100 μL) and a constant amount of thrombin (80 μL, 1 mg mL^{-1}) were performed to study the amount of protein efficiently immobilized on the magnetic nanoparticles. Samples of the supernatants were withdrawn and analyzed for remaining protein by Bradford method and the calibration curve was prepared using gradient concentrations of BSA solutions. The protein immobilized on the magnetic nanoparticles was determined.

Sample preparation

The whole plant of *Erigeron breviscapus* was collected from Dali autonomous prefecture, Yunnan Province, and authorized by Prof. Cai Baochang. Voucher specimens were deposited in the herbarium of Jiangsu Key Laboratory of Chinese Medicine Processing, Nanjing University of Chinese Medicine (no. EB150716). The sample preparation procedures were as follows: first, the whole plant of *Erigeron breviscapus* was pulverized into powder and sieved (60 meshes). Second, an aliquot (1.0 g) of the powder was accurately weighed and refluxed in 50 mL of methanol for 2 h and then filtered. Third, the filtrate was centrifuged at 13 400 rpm for 10 min and the supernatant was stored in −20 °C for the next experiment.

Optimization conditions for magnetic solid-phase extraction of thrombin binders

One well-known compound, *i.e.* scutellarin, was employed to select the best experimental conditions. All experiments were performed in 1.5 mL centrifuge tubes. The reaction mixtures of the model compound (100 μL, 0.1 mg mL^{-1}) and thrombin-immobilized magnetic nanoparticles (50 μL) were prepared in phosphate buffer solution to reach a final volume of 1 mL, and incubated for appropriate time at proper temperature. The supernatant was stored, and the thrombin-immobilized magnetic nanoparticles were washed multiple times with 1 mL of the buffer. The magnetic nanoparticles were subsequently incubated in 1 mL denature solvent for 10 min to dislodge specific bound compounds. The final eluate was collected and retained for analysis *via* UHPLC-Q-TOF-MS.

Several parameters were investigated in the present study to optimize the experimental conditions. At first, varied wash times (1, 2, 3, 4 and 5 times) were investigated for optimization of the washing procedure. Second, different ratios of acetonitrile–water (10, 30, 50, 70 and 90%, v/v) and methanol–water (10, 30, 50, 70 and 90%, v/v; elution step) were investigated for optimization of the denature procedure. Third, incubation temperature (25, 30, 37, 40 and 45 °C) and incubation time (10, 20, 30, 45, 60, 70 and 80 min) were used to study the effect of pH and ion strength on the extraction efficiency. Finally, a gradient pH (5.6, 6.2, 6.8, 7.4 and 8.0) and a gradient concentration (10, 50, 100, 250 and 500 mM) of phosphate buffers were also investigated for optimization of the extraction procedure.

Validation of the established approach

The chromatographic separation was carried out on a reversed-phase XBridge-C_{18} column (250 mm × 4.6 mm, 5 μm, Waters) with the column temperature set at 30 °C. The mobile phase consisted of water containing 0.1% (v/v) formic acid (A) and acetonitrile (B). A gradient program was used according to the following profile: 0 min, 5% B; 25 min, 25% B. The flow rate was 1 mL min^{-1} and the injection volume was 20 μL. The UV spectra were recorded from 190 to 400 nm while the chromatogram was acquired at 254 nm.

The specificity of the method was determined by analyzing thrombin non-binder for interference at the retention times of the thrombin binder. Specificity was assessed by comparing the peak of scutellarin to that in a solution spiked with analyte at 0.1 mg mL^{-1} for scutellarin and specnuezhenide. The magnetic solid-phase extraction procedure was followed as described above. The calibration curves were constructed by plotting the peak area of scutellarin to its gradient concentrations (0.002–0.5 mg mL^{-1}) on the basis of linear regression model. Lower limit of quantification (LLOQ) was defined as the lowest analytical concentration on the calibration curve, which generated S/N ratios of about 10.

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The accuracy and precision of the method were evaluated by repeated analyses of QC samples (0.031 mg mL^{-1} for scutellarin) on three consecutive days. The measured precision was expressed as relative standard deviation (R.S.D.), and the accuracy was expressed as relative error (R.E.). Matrix effects were investigated on crude extract or injection of *Erigeron breviscapus*, by calculating the ratio of the peak area of scutellarin in the presence of crude extract or injection of *Erigeron breviscapus* to the peak area in absence of crude extract or injection of *Erigeron breviscapus* at three different QC concentrations (low, medium, and high).

Applications to crude extract of *Erigeron breviscapus* and its injection

The crude extract and injection of *Erigeron breviscapus* were characterized using ultra-high performance liquid chromatography coupled with Q-TOF 5600-plus mass spectrometer. The acquisition parameters for quadrupole-time-of-flight mass spectrometry were: collision energy, −35 eV; ion spray voltage, −4.5 kV; nebulizer gas (gas 1), 55 psi; declustering potential, −60 V; heater gas (gas 2), 55 psi; turbo spray temperature, 550 °C; and curtain gas, 35 psi. A full MS-IDA-MS/MS scan was acquired on samples, in which a full mass scan was also recorded and the eight most abundant ions were selected for the MS^2 spectra acquisition. The Q-TOF/MS scan range was set as *m*/*z* 100–1500.

The extraction procedure was carried out according to the method described above. The reaction mixtures of the crude extract and injection of *Erigeron breviscapus* (50 μL) and thrombin-immobilized magnetic nanoparticles (50 μL) were prepared in phosphate buffer solution to reach a final volume of 1 mL, and incubated for 70 min at 37 °C using water bath. The supernatant was stored, and the thrombin-immobilized magnetic nanoparticles were washed four times with 1 mL of the buffer. The beads were subsequently incubated in 1 mL of 50% (v/v) acetonitrile/water for 10 min to disclose specific bound compounds. The fifth eluate was collected and retained for analysis *via* UHPLC-Q-TOF-MS.

Thrombin inhibitory assay

Thrombin inhibitory activity was measured according to the method previously described with slight modifications.[14] Substrate solution with a concentration of 0.5 mg mL^{-1} was prepared by sonication of H-D-Phe-Pip-Arg-pNA·2HCl suspended in Tris–HCl buffer (pH 8.2). Samples were dissolved separately in Tris–HCl buffer to yield 0.5 mg mL^{-1} solutions and then diluted to yield gradient concentrations of solutions. After that, the enzyme (90 μL) and sample (20 μL) solutions were added to each well of a 96-well plate and preincubated for 5 min at 37 °C. A substrate solution (90 μL) was added to start the reaction, and the absorbance was monitored at 405 nm before and after 30 min with incubation at 37 °C. Argatroban was used as the positive control.

Results and discussion

Morphological characterization and protein measurement

Transmission electron microscopy image revealed an average diameter of 400–500 nm as shown in Fig. 2A. The close-up of the red dashed line circle in Fig. 2A was present in Fig. 2B. The red arrows clearly shows the presence of thin gray film on the surface of the thrombin conjugated magnetic nanoparticles, indicating thrombin was successfully immobilized onto the surface of magnetic nanoparticles *via* amide reaction. FT-IR analysis indicated a new amide band of 1638 cm^{-1} appeared after the immobilization (see Fig. S1†), which was due to C=O stretching pattern of the secondary amide linkage. This suggested that the conjugation took place in covalent manner through amide linkage. The main peaks of the X-ray diffraction of the magnetic nanoparticles, such as 2.9706, 2.5322, 2.1008, 1.7122, 1.6146 and 1.4837, match well with the standard magnetite Fe_3O_4 XRD spectrum (see Fig. S2†). Magnetization curves of blank magnetic nanoparticles and thrombin-immobilized magnetic nanoparticles are shown in Fig. S3.† Maximum saturation magnetization (57.3 emu g^{-1}) of thrombin-coated magnetic nanoparticles is a little less than that (79.4 emu g^{-1}) of blank magnetic nanoparticles due to the nonmagnetic thrombin on the surface.

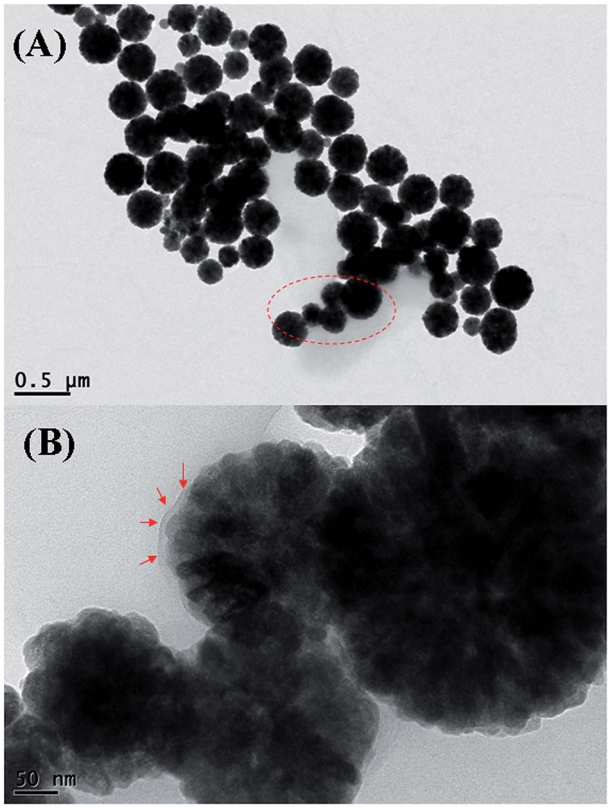


Fig. 2 TEM images of thrombin conjugated magnetic nanoparticles.

The amounts of magnetic nanoparticles were varied to determine the optimal ratio for thrombin binding efficacy with a constant amount of thrombin. As shown in Fig. S4,† with increasing the amount of Fe_3O_4 added at a constant thrombin amount of 80 μg, the amount of thrombin bound increased and then dropped when the amount of Fe_3O_4 added was above 75 μL. The results suggested that a ratio of 75 μL Fe_3O_4 per 80 μg thrombin was sufficient for immobilization. Further increases

in the amount of Fe_3O_4 did not increase the amount of bound thrombin, suggesting that the largest immobilization efficiency had been achieved. The trend was in consistent with a previous report,[15] which used human estrogen receptor α as the binding protein. Aggregation of the Fe_3O_4 was observed when the Fe_3O_4 amount was increased from 75 μL to 100 μL, which may account for the drop in the bound thrombin. Therefore, a ratio of 75 μL Fe_3O_4 per 80 μg thrombin was chosen for subsequent studies.

Optimization conditions for magnetic solid-phase extraction

The washing step is crucial in the magnetic solid-phase extraction because unbound or nonspecific compounds may interfere with the result (see Fig. S5A†). Investigations of different washing times (1–5 times) were carried out. Results suggested that four times of washing was sufficient to remove any unbound compounds. A denaturing solvent was employed to denature the thrombin and dislodge the specifically bound compound. Two-solvent systems (acetonitrile–water and methanol–water), each with five proportions (10, 30, 50, 70 and 90%, v/v), were evaluated. As shown in Fig. S5B,† the extraction efficiency of scutellarin washed by 50% (v/v) acetonitrile–water was larger than that of other solvents. Thus, 50% (v/v) acetonitrile–water was selected as denaturing solvent.

Because the non-covalent interaction between the compound and thrombin is mainly driven by electrostatic interactions,[16] the ionic strength and pH are of great importance to the interaction. Therefore, the effect of pH on the non-covalent interaction was studied in PBS with varied pH values ranging from 5.7 to 8.0. As shown in Fig. S6A,† the increase of pH from 6.8 to 8.0 resulted in an obvious decrease in the bound ratio of scutellarin. The optimal pH for the incubation ranged from 6.2 to 6.8. The isoelectric point (pI) of thrombin is about 7.05, and when the solution pH value is below the pI of thrombin, the thrombin immobilized magnetic nanoparticles have a net positive surface charge, which favors interaction with scutellarin (pK_a 3.29) that have negative charge. The effect of ionic strength was investigated over a broad range of PBS concentrations (10–500 mM). As shown in Fig. S6B,† excessive PBS led to a significant decrease in bound efficiency for thrombin immobilized magnetic nanoparticles, which may due to the charge neutralization on the surface of magnetic nanoparticles. Thus, PBS concentration of 10 mM was chosen in this experiment.

The length of incubation time is a factor affecting the binding degree. As shown in Fig. S6C,† an incubation time of 70 min was sufficient for the interaction between thrombin and scutellarin. Further increase in the incubation time exerted no significant effect on the interaction between thrombin and scutellarin. The incubation temperature was also investigated by varied temperatures ranging from 25 °C to 45 °C. As shown in Fig. S6D,† 37 °C may be the suitable temperature for the incubation due to the largest binding degree at the temperature. Our results were in agreement with the report[17] that the optimal reaction temperature was 37 °C for immobilized thrombin.

In summary, the optimal magnetic solid-phase extraction conditions were as follows: ion strength: 10 mM, pH value: 6.8, incubation time: 70 min, temperature: 37 °C.

Validation of the established approach

Scutellarin is a well known thrombin inhibitor, which belongs to flavone 7-glucuronide.[18] Specnuezhenide, which is an iridoid glycoside, displays negative thrombin inhibitory effect. Therefore, scutellarin and specnuezhenide were selected to investigate the specificity of the magnetic solid-phase extraction method. As was shown in Fig. S7A,† the peak of scutellarin was observed in the presence of specnuezhenide (see black line and red line) after magnetic solid-phase extraction, indicating that the method had good specificity with no interference of the inactive compound specnuezhenide. To avoid the nonspecific binding, denatured thrombin conjugated magnetic nanoparticles was used to be incubated with the mixture of scutellarin and specnuezhenide. None of the unspecific binding was observed (see red line in Fig. S7B†).

As we known, crude extracts of *Erigeron breviscapus* may have many kinds of constituents, including phenolics, flavonoids, fatty acid, polysaccharide and tannins. Tannins are found almost in all plants, which can react with proteins, nitrogenous bases, polysaccharides, some alkaloids, few glycosides and form precipitates. The injection of *Erigeron breviscapus* may have some excipients. The signal from the established approach may be confounded by interference of these nuisance compounds, or by the additive of excipients or synergistic effects of several other compounds. Thus, the aim of this section was to investigate the interference of these nuisance compounds such as tannins or by the additive of excipients or synergistic effects of several other compounds and ensure the result of the approach was reliable.

The calibration curve, linear range, LOD, LOQ, and repeatability of scutellarin were carried out using the developed method. Reasonable correlation coefficient values ($r^2 = 0.9965$) indicated good correlations between scutellarin concentrations and its peak area within the ranges tested. The LOD and LOQ for scutellarin were determined as 0.651 μg mL^{-1} and 1.953 μg mL^{-1}, respectively. The repeatability present as RSD ($n = 6$) was 1.69%. The overall intra- and inter-day variations (RSD) of scutellarin were 2.93% and 1.31%, respectively. The developed method had good accuracy with the recoveries ranging from 95.89% and 107.70% (see Table 1). Taken together, the results demonstrated that the method was sensitive, precise, and accurate enough for magnetic solid-phase extraction of thrombin binders.

Table 1 Recovery of scutellarin in extract and injection of *Erigeron breviscapus*

Sample	Initial concentration (μM)	Added concentration (μM)	Found concentration (μM)	Recovery (%)	RSD (%)
Extract	317.76	67.59	388.80	105.10	3.3
		135.19	460.27	105.42	2.54
		270.38	608.95	107.70	0.48
Injection	865.51	281.19	1135.14	95.89	1.82
		562.38	1453.10	104.48	0.13
		1124.76	2012.45	101.97	1.22

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Applications to crude extract and injection of *Erigeron breviscapus*

The optimized UHPLC-Q-TOF-MS protocol was employed for the separation and identification of compounds in crude extract and injection of *Erigeron breviscapus*. The total ion current chromatograms were shown in Fig. 3A and S8A.† A total of thirty-two compounds were identified, including 15 phenolic acids, 3 flavonoids, 8 unsaturated fatty acid and 6 unknowns, and their detailed information was displayed in Table 2. The retention time and fragmentation information of compounds **3**, **4**, **5**, **16**, **18**, **19**, **20**, **21**, **30**, **31** and **32** were compared with that of standard compounds. Chemical structures of the identified compounds were shown in Fig. S9.†

Phenolic acids have been regarded as the major active constituents in *Erigeron breviscapus*, such as monocaffeoylquinic acids and dicaffeoylquinic acid isomers.[19] Fragmentations with loss of 162 Da (caffeoyl) or production of *m*/*z* 179[caffeic acid − H]$^-$ were the characteristic mass information of these phenolic compounds. The compound corresponding to peak **1** was deduced as protocatechoic acid[20] by its ions at *m*/*z* 153[M − H]$^-$ and *m*/*z* 109[M − H − CO_2]$^-$ in negative MS^2 spectrum (see Table 2). Based on the MS data of peaks **2**, **3**, **4** and **5** (*m*/*z* 353[M − H]$^-$, *m*/*z* 191[M − H − caffeoyl]$^-$, *m*/*z* 179[caffeic acid − H]$^-$ and *m*/*z* 135[caffeic acid − CO_2 − H]$^-$ in negative MS^2 mode, see Fig. S10A†), they were determined to be monocaffeoylquinic acids. Compared with standard compounds, peaks **3**, **4** and **5** were identified as neochlorogenic acid, chlorogenic acid and 4-caffeoylquinic acid, respectively. Peak **2** was deduced as 1-caffeoylquinic acid by comparing its retention time, deprotonated molecules [M − H]$^-$ and characteristic fragments with those reported in the literature.[21] Moreover, peaks **6**, **7** and **8** were determined as caffeic acid derivatives based on the following mass information: *m*/*z* 381[M − H]$^-$, *m*/*z* 219[M − H − caffeoyl]$^-$, *m*/*z* 179[caffeic acid − H]$^-$ and *m*/*z* 161[caffeoyl − H]$^-$ in negative MS^2 mode (see Fig. S10B†). Compared with the ref. 19, peaks **6**, **7** and **8** were tentatively deduced as 3-caffeoyl-2,7-anhydro-3-deoxy-2-octulopyranosonic acids (3-CDOA), 4-CDOA and 9-CDOA, respectively. To the best of our knowledge, 3-CDOA, 4-CDOA and 9-CDOA were only reported in *Erigeron breviscapus*. In addition, peaks **18**, **19** and **20** were determined to be dicaffeoylquinic acids based on the MS data of (*m*/*z* 515[M − H]$^-$, *m*/*z* 353[M − H − caffeoyl]$^-$, 191 [M − H − 2 × caffeoyl]$^-$ and *m*/*z* 179[caffeic acid − H]$^-$ in negative

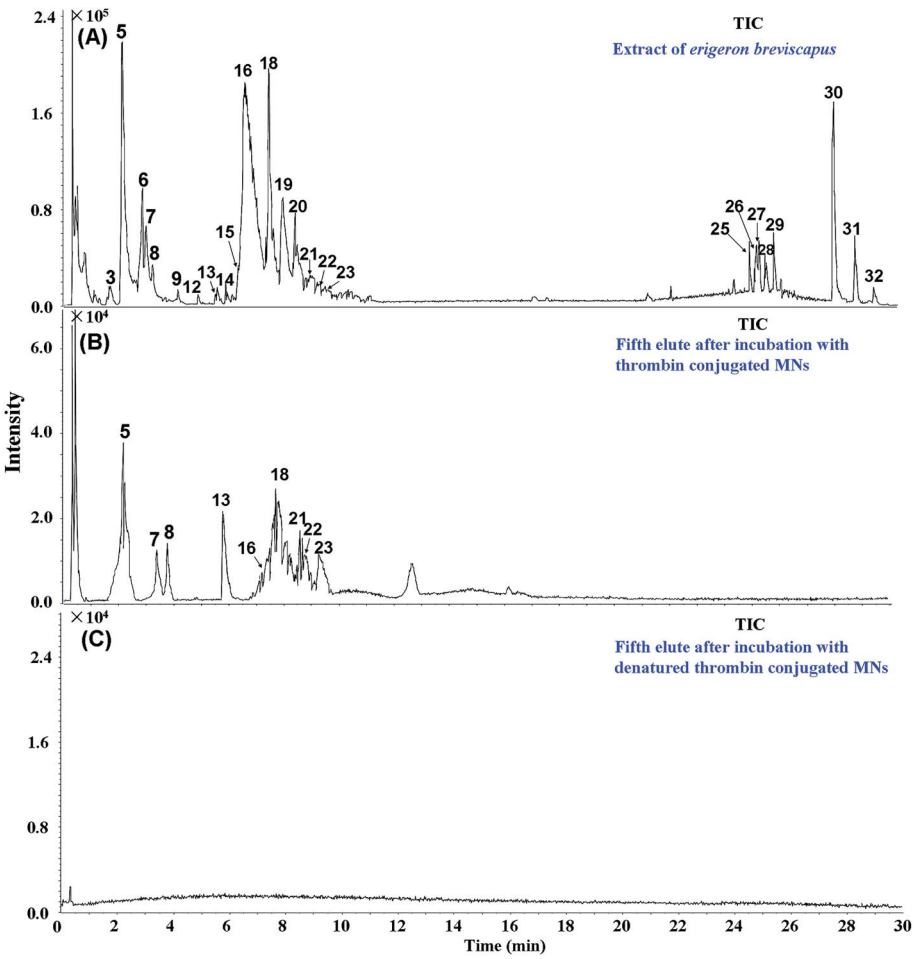


Fig. 3 TIC chromatograms of (A) extract of *Erigeron breviscapus*, (B) fifth eluent after incubation with thrombin immobilized magnetic nanoparticles, (C) fifth eluent after incubation with denatured thrombin immobilized magnetic nanoparticles. Peak numbers are consistent with those in Table 2.

Table 2 MS^1 and MS^2 data of constituents in crude extract and injection of *Erigeron breviscapus* by UHPLC-Q-TOF-MS

No.	t_R (min)	MS^2	Formula	ESI-MS(−) Measured mass $[M-H]^-$	Error (ppm)	Identification	Source
1	1.192	**109.0303**, 108.0224, 91.0197	$C_7H_6O_4$	153.0204	7.0	Protocatechoic acid	Injection
2[a]	1.507	**191.0562**, 179.0349, 135.0450	$C_{16}H_{18}O_9$	353.0869	−2.6	1-Caffeoylquinic acid	Injection
3[a]	1.744	**191.0563**, 179.0350, 135.0452	$C_{16}H_{18}O_9$	353.0863	−4.3	Neochlorogenic acid	Injection, extract
4[a]	1.979	**191.0561**, 179.0348, 135.0448	$C_{16}H_{18}O_9$	353.0865	−3.7	Chlorogenic acid	Injection
5[a]	2.205	**191.0565**, 179.0352, 135.0456	$C_{16}H_{18}O_9$	353.0885	1.1	4-Caffeoylquinic acid	Injection, extract
6	2.717	293.0653, 219.0509, 179.0353, **161.0245**, 135.0453	$C_{17}H_{18}O_{10}$	381.0814	−3.5	3-CDOA	Injection, extract
7	2.925	219.0510, 179.0350, **161.0245**, 135.0451	$C_{17}H_{18}O_{10}$	381.0815	−3.2	4-CDOA	Extract
8	3.289	219.0512, 179.0363, **161.0248**, 135.0458	$C_{17}H_{18}O_{10}$	381.0813	−3.7	9-CDOA	Injection, extract
9	4.153	**387.2002**, 189.1270, 89.0237	$C_{20}H_{34}O_{10}$	433.2080	0.2	Unknown	Extract
10	4.529	251.0569, 191.0558, **179.0353**, 161.0246, 135.0451	$C_{16}H_{18}O_9$	353.0887	2.5	Unknown	Injection
11	4.774	191.0566, 179.0347, **161.0244**, 135.0452	$C_{16}H_{18}O_9$	353.0880	0.5	Unknown	Injection
12	4.883	161.0285, **135.0416**	$C_{13}H_{24}O_8$	307.1403	1.5	Unknown	Extract
13	5.557	**323.0639**, 179.0277, 161.0179	$C_{20}H_{20}O_{11}$	435.0930	1.0	Erigeside I	Extract
14	5.903	**119.0857**	$C_{16}H_{24}O_8$	343.1402	1.0	Unknown	Extract
15	6.303	**287.0558**, 175.0231, 151.0034, 135.0451	$C_{21}H_{20}O_{12}$	463.0858	−5.2	Eriodictyol-7-*O*-glucuronide	Extract
16[a]	6.452	**285.0409**	$C_{21}H_{18}O_{12}$	461.0707	−4.0	Scutellarin	Extract, injection
17	7.467	**301.0361**	$C_{21}H_{18}O_{13}$	477.0674	−0.1	Quercetin-3-*O*-glucuronide	Injection
18[a]	7.816	**353.0881**, 191.0560, 179.0348, 173.0448, 135.0451	$C_{25}H_{24}O_{12}$	515.1165	−5.8	4,5-Di-*O*-caffeoylquinic acid	Extract, injection
19[a]	8.372	**353.0892**, 191.0564, 179.0355, 173.0458	$C_{25}H_{24}O_{12}$	515.1171	−4.7	3,4-Di-*O*-caffeoylquinic acid	Extract, injection
20[a]	8.614	**353.0895**, 191.0561, 179.0354, 173.0461	$C_{25}H_{24}O_{12}$	515.1166	−5.6	3,5-Di-*O*-caffeoylquinic acid	Extract, injection
21[a]	10.386	**381.0829**, 363.0724, 179.0344, 161.0246, 135.0450	$C_{26}H_{24}O_{13}$	543.1119	−4.6	Erigoster B	Extract
22	10.577	**381.0823**, 363.0720, 319.0818, 221.0459, 179.0356, 161.0253, 135.0462	$C_{26}H_{24}O_{13}$	543.1117	−5.0	4,9-Di-CDOA	Extract
23	10.651	**381.0816**, 363.0714, 319.0818, 221.0456, 179.0351, 161.0248, 135.0459	$C_{26}H_{24}O_{13}$	543.1117	−5.0	4,9-Di-CDOA isomer	Extract
24	24.234	**283.0213**	$C_{25}H_{32}O_8$	459.2038	3.0	Unknown	Injection
25	24.850	295.2289, 277.2182, 199.1337, **185.1185**	$C_{18}H_{34}O_4$	313.2398	4.4	12,13-Dihydroxy-9-octadecenoic acid	Extract, injection
26	25.022	**275.2026**, 235.1703, 183.1026	$C_{18}H_{30}O_3$	293.2130	2.7	13-Oxo-9*Z*,11*E*-octadecadienoic acid	Extract
27	25.170	**275.2014**, 195.1397, 183.1392	$C_{18}H_{30}O_3$	293.2131	3.0	9-Oxo-10*E*,12*Z*-octadecadienoic acid	Extract
28	25.374	**273.1871**, 235.1710, 195.1398	$C_{18}H_{28}O_3$	291.1974	2.9	4-Oxo-9,11,13-octadecatrienoic acid	Extract
29	25.746	**277.2187**, 195.1404, 171.1043	$C_{18}H_{32}O_3$	295.2273	−1.9	(9*Z*,11*E*)-13-Hydroxy-9,11-octadecadienoic acid	Extract
30[a]	27.831	**259.2087**	$C_{18}H_{30}O_2$	277.2172	−0.4	Linolenic acid	Extract
31[a]	28.615	**261.2211**	$C_{18}H_{32}O_2$	279.2327	−0.9	Linoleic acid	Extract
32[a]	29.307	**237.2187**	$C_{16}H_{32}O_2$	255.2330	0.2	Hexadecanoic acid	Extract

[a] Compared to standard compounds.

MS^2 mode, see Fig. S10C†). Compared with the standard compounds and related ref. 22, peaks **18**, **19** and **20** were unambiguously identified as 4,5-di-*O*-caffeoylquinic acid, 3,4-di-*O*-caffeoylquinic acid and 3,5-di-*O*-caffeoylquinic acid, respectively. Peaks **21**, **22** and **23** showed the $[M-H]^-$ ion at *m*/*z* 543 and produced ions at *m*/*z* 381$[M-H-\text{caffeoyl}]^-$ and *m*/*z* 363[M − H

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– caffeic acid]$^-$ in MS^2 spectrum (see Fig. S10D†). Peak **21** was unambiguously identified as erigoster B in comparison with the standard compound. The retention time of peaks **22** and **23** were a little later than that of peak **21**. Therefore, peaks **22** and **23** were tentatively deduced as 4,9-di-CDOA and its isomer.[23] Peak **13** showed the [M – H]$^-$ ion at *m*/*z* 435 and produced ions at *m*/*z* 323, *m*/*z* 179[caffeic acid – H]$^-$ and *m*/*z* 161[caffeoyl – H]$^-$ in MS^2 spectrum (see Fig. S11A†). Peak **13** was easily determined to be caffeic acid derivative. Compared with the ref. 4, the compound corresponding to peak **13** was deduced as erigeside I, which was only reported in *Erigeron breviscapus*.

Three flavonoids were identified including peaks **15**, **16** and **17**. The [M – 176 – H]$^-$ ion in the MS^2 spectra of peaks **15**, **16** and **17** corresponded to the presence of glucuronic acid. Peak **15** showed the [M – H]$^-$ ion at *m*/*z* 463 and produced ion at *m*/*z* 287 with a neutral loss of 176 Da in MS^2 spectrum (see Fig. S11B†). The retention time of peak **15** was a little earlier than that of peak **16**. Peak **16** exhibited the [M – H]$^-$ ion at *m*/*z* 461 and gave product ion at *m*/*z* 285 in MS^2 spectrum (see Fig. S11C†). Peak **16** was unambiguously identified as scutellarin by comparison with the standard compound. Compared with the literature,[24] peak **15** was plausibly assigned as eriodictyol-7-*O*-glucuronide. The compound corresponding to peak **17** was deduced as quercetin glucuronide by its ion at *m*/*z* 301[M – glucuronic acid – H]$^-$ in negative MS^2 mode. In comparison with the ref. 24, peak **17** was unambiguously deduced as quercetin-3-*O*-glucuronide.

Eight unsaturated fatty acids were identified in *Erigeron breviscapus*, including peaks **25–32**. Peak **25** showed the [M – H]$^-$ ion at *m*/*z* 313, with the molecular formula of $C_{18}H_{34}O_4$. The unsaturated degree of peak **25** was calculated as two, indicating that no benzene skeleton existed in the molecule. In comparison with the ref. 25, peak **25** was tentatively deduced as 12,13-dihydroxy-9-octadecenoic acid. Similarly, the unsaturated degrees of peaks **26–29** were calculated as four, four, five and three. Compared with the literature,[26,27] peaks **26–29** were assigned as 13-oxo-9*Z*,11*E*-octadecadienoic acid, 9-oxo-10*E*,12*Z*-octadecadienoic acid, 4-oxo-9,11,13-octadeca-trienoic acid and (9*Z*,11*E*)-13-hydroxy-9,11-octadecadienoic acid, respectively. Moreover, peaks **30–32** were unambiguously identified as linolenic acid, linoleic acid and hexadecanoic acid in comparison with standard compounds.

In addition, the concentrations of major compounds in the crude extract and injection of *Erigeron breviscapus* were determined using HPLC-UV and the concentrations of these compounds were listed in Table S1.† The results showed there were small variations between the concentrations of the major compounds in the crude extract and injection of *Erigeron breviscapus*. The concentration of scutellarin in injection was two times larger than that of the extract. Protocatechoic acid, 1-caffeoylquinic acid and chlorogenic acid were only detected in injection, whereas erigoster B was only found in the crude extract.

The developed thrombin immobilized magnetic solid-phase extraction method was applied to the crude extract and injection of *Erigeron breviscapus* and the results were shown (see Fig. 3B, C, S8B and 8C†). A Venn diagram was generated for demonstrating commonly and exclusively detected thrombin binders from the crude extract and injection of *Erigeron breviscapus* (see Fig. 4). Two thirds of the constituents in the injection were detected in the crude extract. Nine thrombin binders were found in the crude extract, whilst seven thrombin binders were identified from the injection. Five compounds, including 4-caffeoylquinic acid (peak **5**), 9-CDOA (peak **8**), erigeside I (peak **13**), scutellarin (peak **16**) and erigoster B (peak **21**) were the common thrombin binders identified.

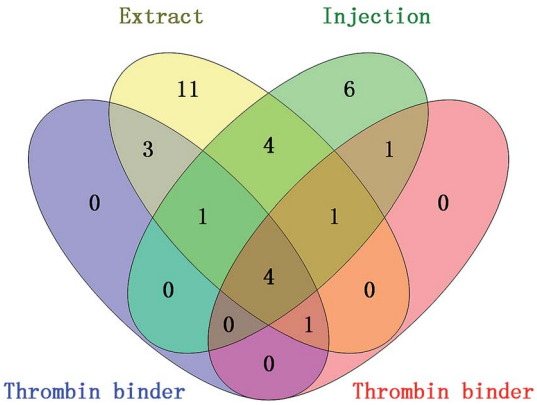


Fig. 4 A Venn diagram demonstrating commonly and exclusively constituents and thrombin binders in crude extract and injection of *Erigeron breviscapus*.

Thrombin inhibitory assay

As shown in Fig. 3B, compounds **5**, **7**, **8**, **13**, **16**, **18**, **21**, **22**, **23** were detected as thrombin binders for the extract. Meanwhile, compounds **4**, **5**, **8**, **13**, **16**, **19**, **21** were detected as thrombin binders for the injection as present in Fig. S8B.† Collectively, compounds **4**, **5**, **7**, **8**, **13**, **16**, **18**, **19**, **21**, **22**, **23** should be evaluated for their inhibitory effect against thrombin. However, due to the lack of standard compounds, the inhibitory effects **7**, **8**, **13**, **22** and **23** were not well elucidated at present. In the future, their inhibitory effects will undoubtedly be elucidated with the endeavor of phytochemistry research. Therefore, fourteen compounds were obtained and evaluated for their inhibitory activities against thrombin using thrombin inhibitory assay. Argatroban served as the positive control. The IC_{50} value of argatroban was determined to be 0.935 μM, which was in agreement with report in the ref. 23. The assay results of other compounds were listed in Table 3. Erigoster B, which was only identified in *Erigeron breviscapus*, exhibited the best thrombin inhibitory activity with an IC_{50} value of 37 μM. Scutellarin showed a little weaker inhibitory effect than that of erigoster B. Compounds **3–5** and **18–20** were caffeoylquinic acids derivatives. Their structures were shown in Fig. S9.† The only difference between them was the number of substitution group (one or two caffeoyl group) and the location site of the caffeoyl group. The thrombin inhibitory activity of them were decreased in the following order: 4,5-di-*O*-caffeoylquinic acid, chlorogenic acid, neochlorogenic acid, 3,5-di-*O*-caffeoylquinic acid, 4-caffeoylquinic acid, 3,4-di-*O*-caffeoylquinic acid. As we expected, unsaturated fatty acids, including linolenic acid, linoleic acid and hexadecanoic acid, showed less than 10% inhibition against thrombin at the concentration of 200 μM. Due to the

Table 3 Inhibitory effects of individual compound on thrombin activities

No.	Compound	IC_{50} (μM) ± SD
3	Neochlorogenic acid	70.645 ± 2.314
4	Chlorogenic acid	63.290 ± 3.493
5	4-Caffeoylquinic acid	81.864 ± 0.770
16	Scutellarin	42.534 ± 8.491
18	4,5-Di-*O*-caffeoylquinic acid	62.431 ± 0.538
19	3,4-Di-*O*-caffeoylquinic acid	92.003 ± 2.002
20	3,5-Di-*O*-caffeoylquinic acid	79.329 ± 3.613
21	Erigoster B	36.966 ± 2.622
30	Linolenic acid	ni[a]
31	Linoleic acid	ni[a]
32	Hexadecanoic acid	ni[a]
	Erigeroside	104.760 ± 3.038
	Scutellarin A	90.700 ± 0.236
	Luteolin 7-*O*-β-D-glucuronide	94.509 ± 1.201
Control	Argatroban	0.935 ± 0.077

[a] Less than 10% inhibition at 200 μM.

unavailability of 9-CDOA and erigeside I in commercial, erigeroside, scutellarin A and luteolin 7-*O*-β-D-glucuronide, which had been reported in *Erigeron breviscapus*, were purchased and tested for their thrombin inhibitory activity. The IC_{50} values of erigeroside, scutellarin A and luteolin 7-*O*-β-D-glucuronide were determined to be 104.8 μM, 90.7 μM and 94.5 μM, respectively. Collectively, eleven compounds showed thrombin inhibitory activity, ten of which were reported for the first time. In summary, new thrombin inhibitory compounds were rapidly discovered from natural medicines using the established approach.

Conclusions

In the study, we have demonstrated a new magnetic solid-phase extraction approach coupled with UHPLC-Q-TOF-MS for identifying thrombin binders from crude extract and injection of *Erigeron breviscapus*. The new established approach provides an alternative to the traditional methods for discovering antithrombin agents from the medicinal plants and its preparations. Particularly, eleven thrombin inhibitors were identified, ten of which were reported for the first time.

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References

1 J. I. Borissoff, H. M. H. Spronk, S. Heeneman and H. ten Cate, *Cardiovasc. Res.*, 2009, **82**, 392–403.
2 J. I. Borissoff, H. M. H. Spronk and H. ten Cate, *N. Engl. J. Med.*, 2011, **364**, 1746–1760.
3 M. Gao, W. Huang, M. RoyChowdhury and C. Liu, *Anal. Chim. Acta*, 2007, **591**, 161–166.
4 Y. Zhang, Q. Zhao, J. Ma, B. Wu and X. Zeng, *Chromatographia*, 2010, **72**, 651–658.
5 S. Wang, H. Wang, H. Guo, L. Kang, X. Gao and L. Hu, *Neuroscience*, 2011, **185**, 150–160.
6 X. M. Hu, M. M. Zhou and F. D. Zeng, *Acta Pharmacol. Sin.*, 2005, **26**, 1454–1459.
7 E. D. Bodnar and H. Perreault, *Anal. Chem.*, 2013, **85**, 10895–10903.
8 M. J. McFadden, M. S. Junop and J. D. Brennan, *Anal. Chem.*, 2010, **82**, 9850–9857.
9 L. Yi, Y. Huang, T. Wu and J. Wu, *Neural Regener. Res.*, 2013, **8**, 3036–3046.
10 B. Chung, G. W. Shin, W. Choi, J. Joo, S. Jeon and G. Y. Jung, *Electrophoresis*, 2014, **35**, 3283–3289.
11 H. Xu, Z. P. Aguilar, L. Yang, M. Kuang, H. Duan, Y. Xiong, H. Wei and A. Wang, *Biomaterials*, 2011, **32**, 9758–9765.
12 S. H. Huang, M. H. Liao and D. H. Chen, *Biotechnol. Prog.*, 2003, **19**, 1095–1100.
13 Y. Tao, Y. Zhang, Y. Cheng and Y. Wang, *Biomed. Chromatogr.*, 2013, **27**, 148–155.
14 A. R. Anas, T. Kisugi, T. Umezawa, F. Matsuda, M. R. Campitelli, R. J. Quinn and T. Okino, *J. Nat. Prod.*, 2012, **75**, 1546–1552.
15 N. Jonker, A. Kretschmer, J. Kool, A. Fernandez, D. Kloos, J. G. Krabbe, H. Lingeman and H. Irth, *Anal. Chem.*, 2009, **81**, 4263–4270.
16 A. Y. Mehta, Y. Jin and U. R. Desai, *Expert Opin. Ther. Pat.*, 2014, **24**, 47–67.
17 G. Wu, X. Deng, X. Li, X. Wang, S. Wang and H. Xu, *Appl. Microbiol. Biotechnol.*, 2011, **92**, 85–93.
18 X. Tian, L. Chang, G. Ma, T. Wang, M. Lv, Z. Wang, L. Chen, Y. Wang, X. Gao and Y. Zhu, *Biol. Pharm. Bull.*, 2016, **39**, 181–191.
19 Y. Zhang, P. Shi, H. Qu and Y. Cheng, *Rapid Commun. Mass Spectrom.*, 2007, **21**, 2971–2984.
20 Q. Chu, T. Wu, L. Fu and J. Ye, *J. Pharm. Biomed. Anal.*, 2005, **37**, 535–541.
21 S. G. Liao, L. J. Zhang, C. B. Li, Y. Y. Lan, A. M. Wang, Y. Huang, L. Zhen, X. Z. Fu, W. Zhou, X. L. Qi, Z. Z. Guan and Y. L. Wang, *Rapid Commun. Mass Spectrom.*, 2010, **24**, 2533–2541.
22 J. Li and D. Yu, *Zhongguo Zhongyao Zazhi*, 2011, **36**, 1458–1462.
23 J. Zhao, S. I. Khan, M. Wang, Y. Vasquez, M. H. Yang, B. Avula, Y. H. Wang, C. Avonto, T. J. Smillie and I. A. Khan, *J. Nat. Prod.*, 2014, **77**, 509–515.
24 J. Li and D. Q. Yu, *J. Asian Nat. Prod. Res.*, 2013, **15**, 969–973.
25 C. T. Hou, *Asia Pac. J. Clin. Nutr.*, 2008, **17**(1), 192–195.
26 I. Lajter, S. P. Pan, S. Nikles, S. Ortmann, A. Vasas, B. Csupor-Loffler, P. Forgo, J. Hohmann and R. Bauer, *Planta Med.*, 2015, **81**, 1270–1276.
27 I. Martin-Arjol, M. Bassas-Galia, E. Bermudo, F. Garcia and A. Manresa, *Chem. Phys. Lipids*, 2010, **163**, 341–346.

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